Week 5 Reference Sheet: Conditional Logic

CSC-125-201 | Use this as a reference when completing Homework 5

Homework Topics

- Basic `if` statements
- `if-else` two-way decisions
- `if-elif-else` chains
- Logical operators: `and`, `or`, `not`
- Combining conditions
- Nested conditionals
- Indentation rules
- Common conditional errors

Basic if Statement

Runs the indented block only when the condition is `True`.

```
temperature = 95
if temperature > 90:
    print("It is hot outside")
print("Have a good day")  # Always runs
```

if-else Statement

Two branches. Exactly one block always runs.

```
number = 7
if number % 2 == 0:
    print(f"{number} is even")
else:
    print(f"{number} is odd")
```

if-elif-else Chain

Multiple conditions checked top to bottom. First `True` wins, rest are skipped.

```
score = 85
if score ≥ 90:
    grade = "A"
elif score ≥ 80:
    grade = "B"
elif score ≥ 70:
    grade = "C"
elif score ≥ 60:
    grade = "D"
else:
    grade = "F"
print(f"Your grade is {grade}")
```

Logical Operators

Operator	Meaning	Example	Result

`and`	Both must be `True`	`True and False`	`False`
`or`	At least one `True`	`True or False`	`True`
`not`	Reverses the value	`not True`	`False`

Truth Table

A	B	A and B	A or B	not A
`True`	`True`	`True`	`True`	`False`
`True`	`False`	`False`	`True`	`False`
`False`	`True`	`False`	`True`	`True`
`False`	`False`	`False`	`False`	`True`

Combining Conditions

```
age = 25
has_license = True
if age >= 16 and has_license:
    print("You can drive")

day = "Saturday"
if day == "Saturday" or day == "Sunday":
    print("It is the weekend")

logged_in = False
if not logged_in:
    print("Please log in")
```

Nested Conditionals

Place an `if` inside another `if`. Each level adds 4 more spaces.

```
has_ticket = True
age = 15
if has_ticket:
    if age >= 18:
        print("Welcome to the R-rated movie")
    else:
        print("You need a parent for this movie")
else:
    print("Please purchase a ticket first")
```

Indentation Rules

- Use **4 spaces** per level (not tabs)
- A **colon** is required after every `if`, `elif`, and `else`
- All lines in the same block must have the **same indentation**

Common Errors

Error	Cause	Fix
`SyntaxError`	Missing colon after condition	Add `:` at end of `if` / `elif` / `else`
`IndentationError`	Inconsistent spacing	Use 4 spaces consistently per level
Logic error	Using `=` instead of `==`	`=` assigns, `==` compares
Logic error	Wrong `elif` order	Check highest threshold first

Resources

- Corey Schafer - Python Conditionals and Booleans
- W3Schools - Python If...Else
- Real Python - Conditional Statements